



Pendular Oscillation and Ocular Bobbing After Pontine Hemorrhage

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Abstract

The pathophysiology of acute, vertical spontaneous eye movements following pontine hemorrhage is not well understood. Here, we present and discuss the video-oculography findings of a patient with acute pontine hemorrhage who developed vertical pendular oscillation and ocular bobbing while comatose. The amplitudes, peak velocities, frequency distribution, and phase planes (velocity versus position) of the eye movements were analyzed. The vertical pendular oscillation was rhythmic with a peak frequency of 1.7 Hz, but amplitudes (mean 1.9°, range 0.2–8.2°) and peak velocities (mean 20.6°/s; range 5.9–60.6°/sec) fluctuated. Overall, their peak velocities were asymmetric, faster with downward than upward. Higher peak velocities were seen with larger amplitudes (downward phase $r = 0.95$, $p < 0.001$; upward phase $r = 0.91$, $p < 0.001$) and with movements beginning at eye positions lower in the orbit (downward phase $r = -0.64$, $p < 0.001$; upward phase $r = -0.86$, $p < 0.001$). Interspersed were typical ocular bobbing waveforms with a fast (peak velocity 128.8°/s), large-amplitude (17.5°) downward movement, sometimes followed by a flat interphase interval (0.5 s) when the eye was nearly stationary, and then a slow return to mid-position with a decaying velocity waveform. To account for the presence and co-existence of pendular oscillations and bobbing, we present and discuss three hypothetical models, not necessarily mutually exclusive: (1) oscillations originating in the inferior olives due to disruption of the central tegmental tract(s); (2) unstable neural integrator function due to pontine cell group damage involving neurons involved in gaze-holding; (3) low-frequency saccadic intrusions following omnipause neuron damage.

Keywords Pendular oscillation · Ocular bobbing · Nystagmus · Pontine hemorrhage · Inferior olive

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Background

Pontine hemorrhage can cause various forms of nystagmus and ocular oscillations, although one of the most characteristic (and localizing) is considered “typical” ocular bobbing. Ocular bobbing was first described by C. Miller Fisher [1] as an intermittent fast downward excursion of the eyes followed by a slow return to center eye position. Typical ocular bobbing has four main features: an irregular rate, a faster downward movement than upward movement, a quiet inter-phase interval, and a large amplitude downward movement [1, 2]. Although rare, when this typical pattern of ocular bobbing is seen, it usually is in comatose patients with absent horizontal eye movements (damage to the abducens nucleus and/or fascicle and/or the medial longitudinal fasciculus) due to acute pontine hemorrhage. The mechanism was initially thought to be a release of the (spared) vertical eye movements following total paralysis of horizontal gaze [1]. Since that time, other patterns of “atypical” bobbing (i.e., anything other than the pattern described above) have been described, including ocular dipping or inverse bobbing (initial movement is slow and downward), reverse bobbing (initial movement is fast and upward), converse bobbing (initial movement is slow and up), and pretectal V-pattern pseudo bobbing (initial movement is fast and down and convergent) [2–4]. While typical bobbing is usually due to pontine hemorrhage (less commonly ischemia or external pontine compression), atypical bobbing has many causes, including metabolic encephalopathy, cerebral hypoxia, and encephalitis [2].

Vertical pendular oscillations have also been described in patients with acute pontine hemorrhage [5, 6]. This was initially referred to as “ocular myoclonus” and was thought to be a variant of oculopalatal tremor (OPT) [6]. However, the appearance of vertical oscillations in the acute phase of stroke is distinct from typical OPT, when vertical and/or torsional pendular nystagmus and palatal tremor develop weeks, months, or even years after the insult [6, 7]. Furthermore, while a pendular oscillation can be present acutely, palatal tremor or other synchronous movements involving branchial arch musculature develop later [5, 6]. Because this pendular-like oscillation of the eyes has asymmetric velocities (i.e., faster downward movements), it can be mistaken for ocular bobbing. It remains unclear whether bobbing, (acute) vertical pendular oscillations, and OPT exist on a spectrum or whether they arise from distinct mechanisms.

Previous reports of ocular bobbing and acute vertical pendular oscillations are mostly descriptive, with oculographic records either lacking or limited. Here, we present video-oculography (VOG) recordings and analysis in a patient who had abnormal vertical eye movements immediately following a pontine hemorrhage. The VOG showed two distinct types of movements: (1) a vertical pendular oscillation with small-to-medium amplitudes, and (2) a superimposed

intermittent large-amplitude waveform that was characteristic of ocular bobbing. We quantified the pattern of oscillations to help determine their pathophysiology.

Methods

Case

A 49-year-old man suffered a pontine hemorrhage causing coma and quadriplegia. On admission to the ICU, he was unresponsive, had pinpoint pupils, and there were no horizontal eye movements, suggesting a pontine localization. He had spontaneous vertical eye movements which appeared to be a combination of smaller amplitude continuous vertical pendular oscillations with intermixed larger amplitude fast downward eye movements consistent with typical ocular bobbing. The amplitude of the vertical eye movements lessened after 2 weeks (video 1). No palatal tremor was noted either at the initial examination or at a follow-up visit.

VOG Recording and Analysis

This patient’s spontaneous eye movement was recorded at the bedside on the second day of hospitalization with a portable VOG device (EyeSeeCam, Interacoustics, Germany; frame rate 220 fps). The raw eye movement data was analyzed and results plotted in MatLab (v2018b). Savitzky-Golay filtering was used for plotting velocity traces. For the pendular oscillations, the frequency distribution and the phase plane (velocity versus position) were analyzed. Means and standard deviations of the downward and upward peak velocities were calculated and compared with Student’s *t* test. The amplitudes of oscillation, starting eye positions, and peak velocities were correlated using Pearson’s correlation coefficient. Statistical significance was assessed with SPSS (version 23). For the larger amplitude bobbing movements, durations and peak velocities were calculated individually. Intervals when the oscillations ceased were also identified and their duration determined.

Results

Figure 1 (video 2) shows a 56-s VOG record. There are continuous vertical pendular oscillations throughout the record. Although the pendular oscillation is rhythmic with a peak frequency of 1.7 Hz (Fig. 2c), the amplitudes (mean 1.9°, range 0.2–8.2°) and peak velocities (mean 20.6°/s; range 5.9–60.6°/s) fluctuate. The recording is also intermixed with faster and larger amplitude downward movements at 11, 14, 19, 22, and 26 s (arrows), compatible with the bobbing seen in the video (video 2).

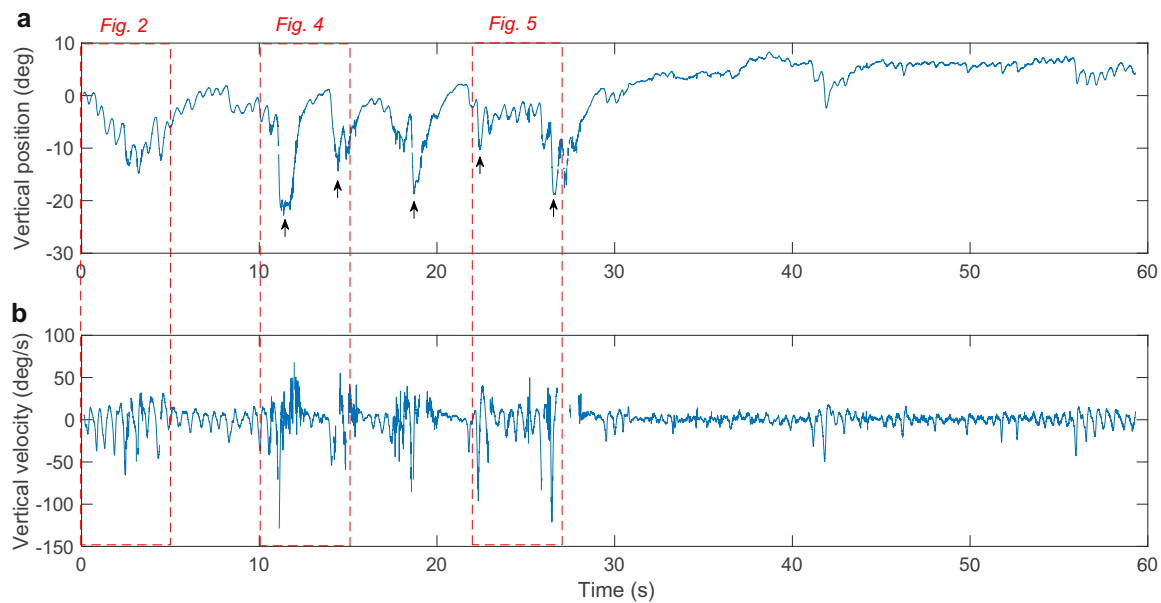


Fig. 1 Vertical pendular oscillation and ocular bobbing in 56-s VOG record. A. Position trace. B. Velocity trace. Fig. 2 (A, B), Fig. 4, and Fig. 5 are amplified from the segments of Fig. 1, labelled with the red

dash frames. (positive deflection in Y-axis: upward movement; negative deflection in Y-axis: downward movement)

Figure 2 a and b shows a 5-s VOG sample demonstrating the characteristics of the pendular oscillation. The peak velocity is asymmetric, faster with downward than with upward movements (downward $22.9 \pm 12.1^\circ/\text{s}$; upward $18.3 \pm 10.0^\circ/\text{s}$ [mean $\pm$ SD], $p = 0.025$). Although the waveform appears almost pendular in the position trace, its sawtooth appearance is clearly seen in the velocity trace, with a rounded positive (upward) velocity turnaround (labelled as *) and pointed negative (downward) velocity turnaround (labelled as an arrow) (Fig. 2b). Another way to clearly see this asymmetry is with a phase-plane diagram (Fig. 2d). Note that the upward movements (red) accelerate more quickly to their peak velocity than the downward movements (black), which accelerate more slowly but eventually reach a higher peak velocity than the upward movements. Once the upward movements (red) begin to slow down, they decelerate more gently than the downward movements (black). Another way to describe the difference between up and down movements is that one is the mirror reversal of the other.

Figure 3 shows additional analyses of amplitudes, eye positions, and peak velocities of the vertical pendular oscillations. The amplitudes vary from 0.2 to 8.2° in downward movements and from 0.5 to 7.3° in upward movements. There is a positive correlation between amplitude and peak velocity both in the downward phase ($r = 0.95$, $p < 0.001$, Fig. 3a) and upward phase ($r = 0.91$, $p < 0.001$, Fig. 3b)—larger peak velocities are seen with larger amplitudes. There is a negative correlation between starting eye positions and peak velocities in downward ($r = -0.64$, $p < 0.001$, Fig. 3c) and upward phases ($r = -0.86$, $p < 0.001$, Fig. 3d)—regardless of the initial

direction of the movement (up or down), the lower the initial position of the eye, the faster it moves.

Figure 4 a and b show a large downward movement beginning at the 11th second of recording. The amplitude is 17.5° , with a downward peak velocity of $128.8^\circ/\text{s}$ (Fig. 4b). The duration of the biphasic wave is 1.9 s (from the 10.9 seconds [upward-downward turning point, labelled as the first *] to the 12.8 seconds [upward-downward turning point, labelled as the second *] of record), which is followed by an interval where there is marked suppression of the lower-amplitude oscillations for about 0.5 s (from the 11.2 to the 11.7 seconds of record), followed by a slow drift upward, and then further suppression until the oscillation begins again about 1.1 s after the initial large downward movement (from the 13.9 seconds of record).

Figure 5 a and b show additional fast downward movements at 22, 26, and 26.5 s. Although the downward peak velocities are greater than $50^\circ/\text{s}$, there was no tonic interphase intervals, and their durations (0.5, 0.5, and 0.6 s) were shorter than the larger amplitude down movements characteristic of ocular bobbing.

Discussion

The relationship between typical ocular bobbing and vertical pendular oscillations, especially in the acute phase of brainstem stroke, has not been well documented since their original descriptions, lacked quantitative analyses, [1–3, 8]. In the last few decades, there have only been a few case reports and video recordings for atypical

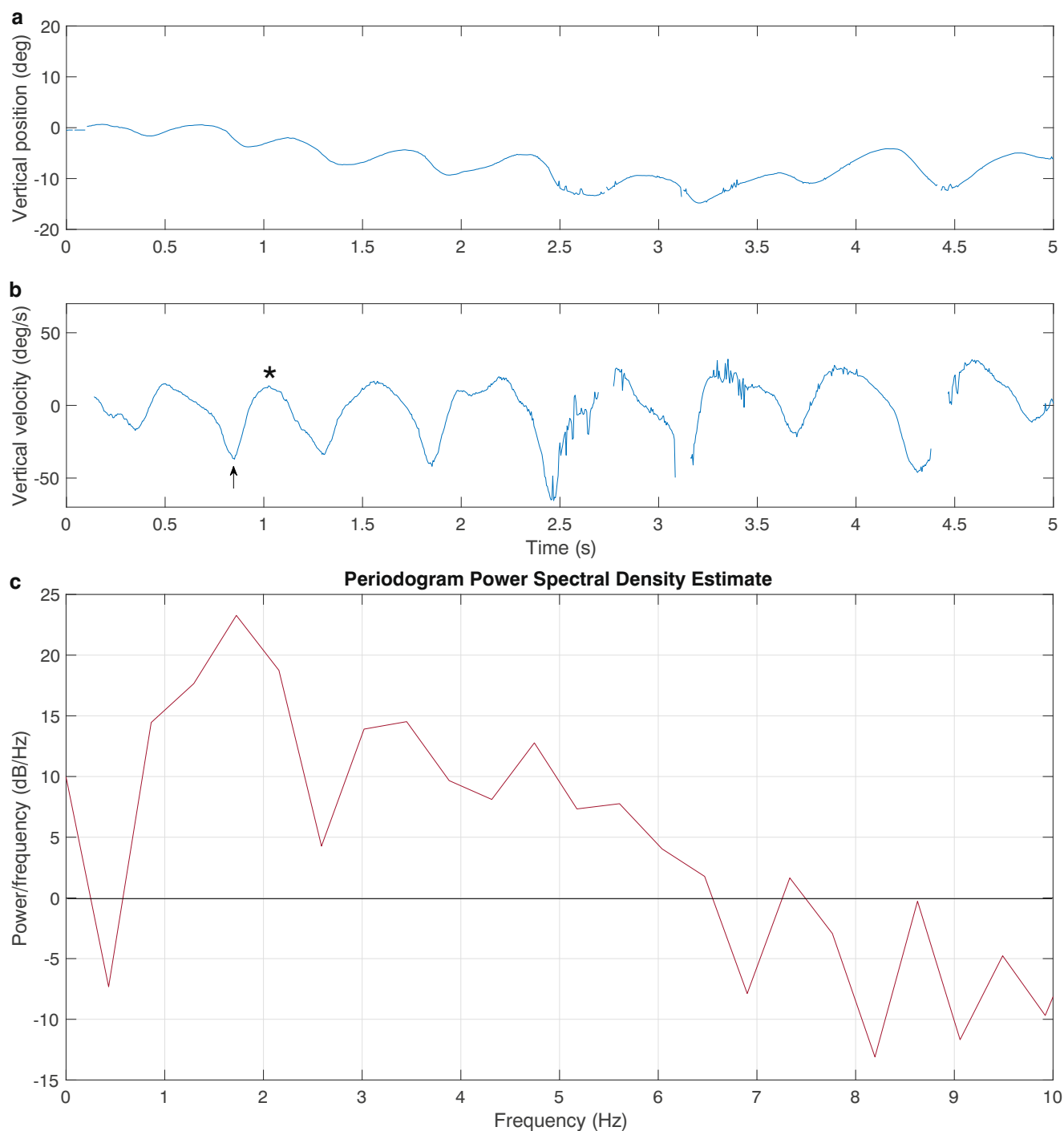


Fig. 2 VOG, power spectrum, and phase plane analysis of the vertical pendular oscillation. **a** Position trace of the 5-s sample of pendular oscillation (blown up from Fig. 1). **b** Velocity trace of the 5-second sample of pendular oscillation (asterisk: rounded turnaround; arrow: pointed turnaround). **c** Power spectrum of pendular oscillation. **d** Phase

plane plot (velocity versus position) of pendular oscillation (thin black lines: individual downward phases; thin red lines: individual upward phases; thick black line: regression line of downward phases; thick red line: regression line of upward phases). Traces are normalized for comparison

bobbing [9–13]. With regard to papers discussing vertical pendular oscillations following an acute pontine lesion, we only found three case reports published more than 20 years ago [5, 6, 14], with one electro-oculographic record in a conscious patient [5]. Our report is the first VOG

analysis in a comatose patient due to an acute pontine hemorrhage who had both vertical pendular oscillations and superimposed ocular bobbing movements.

The VOG records and quantitative analysis of the oscillations shown by our patient uncovered a highly complex

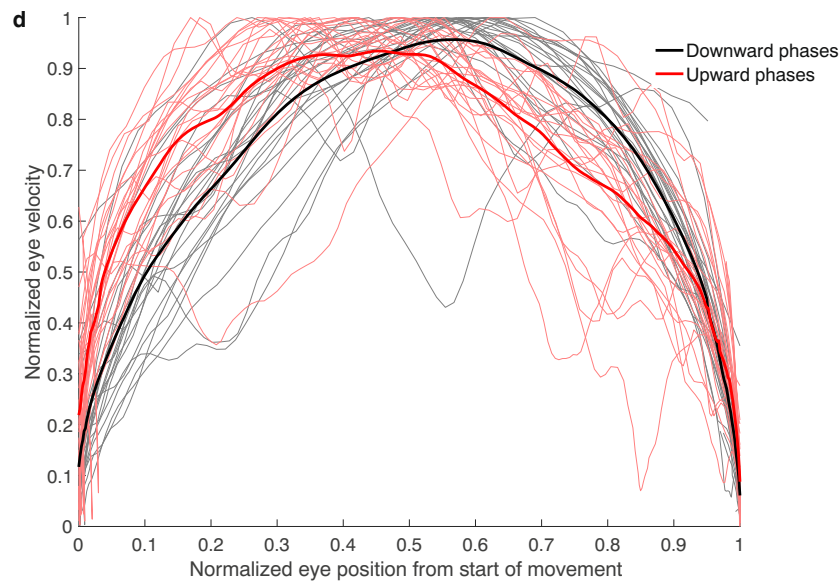


Fig. 2 continued.

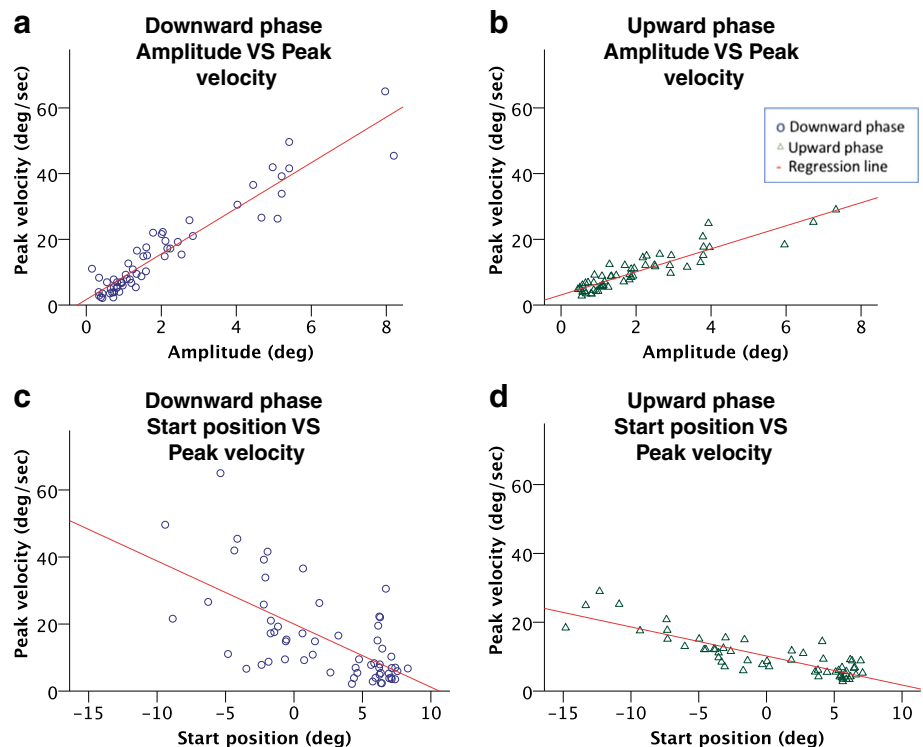
disorder of eye movements. We were able to quantify these complex oscillations using measures of dynamic properties, such as peak velocity, as well as phase plane analyses to compare their trajectories more precisely. While an immediate, simple underlying patho-physiology did not emerge from our analyses, our findings allowed us to explore potential explanations based upon recent concepts of the anatomy and physiology that underlie the control of normal eye movements by circuits in the brainstem and cerebellum. We will present

three hypotheses for consideration, and speculate how they might explain the dramatic oscillations that occur in the setting of acute dysfunction in the pons.

The Acute Olivary Oscillation Hypothesis

Previously, the mechanism(s) for (acute) vertical pendular oscillations and OPT following pontine hemorrhage were thought to be the same [6]. However, the pathophysiology of

Fig. 3 Distribution plots among amplitude, starting position of the eye and peak velocity in vertical pendular oscillations. **a** Downward amplitude vs. peak velocity. **b** Upward amplitude vs. peak velocity. **c** Start position of the eye to go downward vs. peak velocity. **d** Start position of the eye to go upward vs. peak velocity



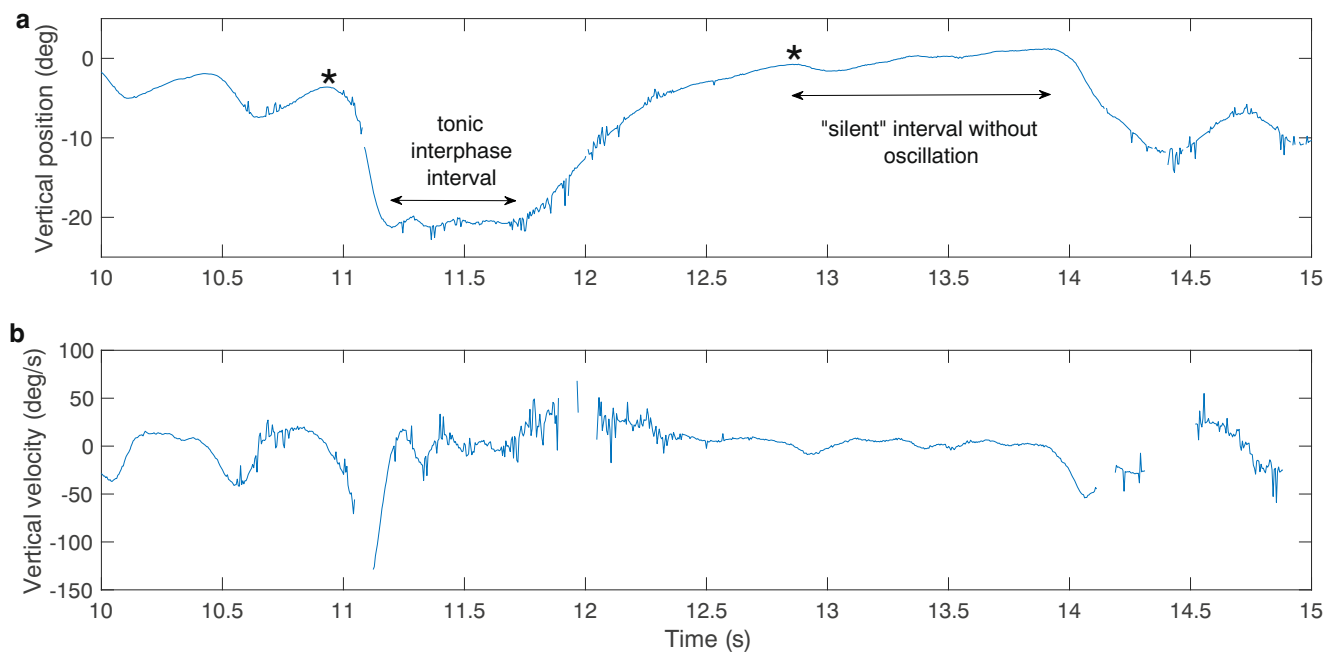


Fig. 4 A cycle with a large amplitude movement compatible with ocular bobbing (blown up from Fig. 1). **a** Position trace. **b** Velocity trace. (1st asterisk: start point of the cycle; 2nd asterisk: endpoint of the cycle)

OPT has been attributed to a delayed effect from chronic deafferentation of the inferior olive leading to hypertrophic olivary degeneration [7, 15–17]. A pontine hemorrhage that involves the central tegmental tract (CTT) is a common cause of OPT, but the oscillations usually develop weeks or months after the insult. The CTT is part of Guillain-Mollaret triangle, consisting of connections between the inferior olive to the

contralateral dentate nucleus; dentate to the contralateral red nucleus where fibers wrap around the red nucleus (but do not synapse there) and descend as the CTT to the ipsilateral inferior olive. Injury to the CTT results in inferior olivary denervation and disinhibition. The characteristic hypertrophic olivary degeneration takes time to develop as the neurons in the denervated inferior olivary nucleus enlarge and form new

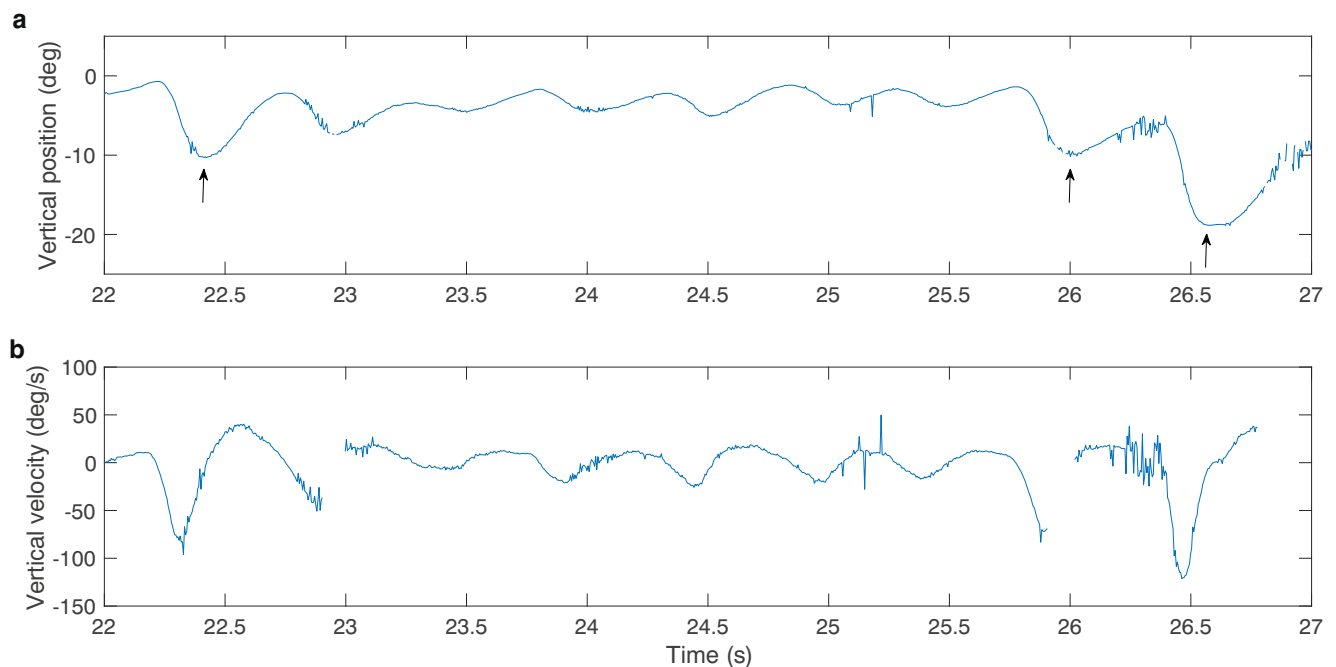


Fig. 5 Large “transitional” waves with features of both ocular bobbing and pendular oscillations (blown up from Fig. 1). **a** Position trace. **b** Velocity trace (arrows: large waves)

gap junctions between the enlarging cell bodies. As gap junctions form, groups of neurons synchronize their firing and these discharges travel to the cerebellum where they are further modulated. This hypothesis provides a plausible explanation for the delay in developing OPT following a pontine hemorrhage but does not explain the appearance of pendular oscillations simultaneous with or immediately following pontine stroke, as in our case.

In one case series, 4 of the 7 patients developed palatal tremor or other branchial rhythmic movements 2–9 months after the (acute) pendular oscillations appeared [6]. Similar findings were noted in two other reports [5, 14]. In OPT, the pathophysiology underlying pendular nystagmus and palatal tremor results from a lesion that disrupts Guillain-Mollaret triangle. However, the development of hypertrophic olivary degeneration may not be necessary for the initiation of vertical oscillations, as evidenced by their occasional presence in the acute setting. In animal studies, Llinas et al. demonstrated that neurons of inferior olives produce oscillatory discharges at various frequencies, and synchronized oscillatory discharges can be induced by some chemicals (e.g., hamarline), toxins (e.g., tetrodotoxins), or neurotransmitters (e.g., serotonin) [18]. Long et al. demonstrated that hamarline-induced tremors were both robust and indistinguishable in mice with strong or weak electrical couplings by gap junctions [19]. These findings suggest that a synchronized oscillatory response from olivary neurons can be triggered immediately and does not require the formation of new gap junctions (i.e., a process that normally takes weeks to months).

The Unstable Neural Integrator Theory

While injury to the CTT is a common cause of OPT, it has been suggested that disruption of the pontine cell groups contributing to gaze-holding networks may also cause pendular nystagmus, usually in the setting of multiple sclerosis (MS) [20, 21]. For example, the cell groups of the paramedian tract (PMT) have excitatory projections to the cerebellar flocculus, which, in turn, helps to optimize function of the brainstem neural integrators (predominantly the nucleus prepositus hypoglossi (NPH) and medial vestibular nucleus (MVN) horizontally and interstitial nucleus of Cajal [INC] vertically) and allow for steady gaze holding when the eyes maintain an eccentric position.

Injury to the PMT or the perihypoglossal nuclei might cause an unstable neural integrator and nystagmus with increasing slow-phase velocity waveforms which violates Alexander's law (e.g., upbeat nystagmus that increases in downgaze) [20]. Likewise, an unstable neural integrator might also result in pendular nystagmus [21]. In the quantitative analysis of our case, the faster downward movements at lower eye positions violates Alexander's law and implies an unstable neural integrator. Acquired pendular nystagmus due to an unstable neural integrator can be transiently suppressed by

saccades, rapid vestibular eye movement (vestibulo-ocular reflex), or blinks [20, 21]. In our case, the continuous pendular oscillations are transiently suppressed by the intermittent ocular bobbing movements, which might imply an unstable neural integrator as well. However, the pendular nystagmus associated with pontine demyelinating lesions is usually horizontal, has a small amplitude (1°), low mean peak velocity ($6^\circ/\text{s}$), high frequency (4–6 Hz), and has a more regular sinusoidal waveform [22]. In contrast, nystagmus from OPT is often more vertical/torsional than horizontal (though this may partially reflect the commonly associated horizontal gaze paresis), has a large mean amplitude (8°), higher peak velocity ($16^\circ/\text{s}$), lower frequency (2–3 Hz), and irregular waveforms [22, 23]. In our case, the vector is vertical, with a mean amplitude 1.9° , mean peak velocity $20.6^\circ/\text{s}$, frequency 1.7 Hz, and has an irregular waveform. Therefore, the movements seen in our patient are more reminiscent of OPT than what occurs with pontine demyelinating lesions.

The Slow Saccadic Intrusion Hypothesis

Our VOG record at the 11th second shows a large-amplitude high-velocity downward movement from a central eye position followed by a tonic, quiet, interphase interval and a subsequent slow return to central position (Fig. 4). This waveform is compatible with the typical ocular bobbing that has been previously reported (i.e., large amplitude, faster downward phase, and presence of a tonic interphase interval) [2] with the proviso that the speed of the downward movement is still slower than that of vertical saccades in normal subjects. Therefore, we provide evidence that acute pendular oscillations and ocular bobbing can co-exist in the acute phase of a pontine hemorrhage, a finding that has previously been observed at the bedside [6].

Notably, although the continuous oscillations in our case show a pendular waveform, the downward movements are faster than upward movements. This is consistent with the findings in brainstem hemorrhage of Lawrence et al., where it was shown that downward velocity exceeded the preceding or subsequent upward velocity by a factor of 1.5 to 4 though the difference in peak velocities for a given amplitude in our patient was in the bottom part of that range [5]. Whether the onset of acute pendular oscillations and ocular bobbing is simultaneous (as in our case), or sequential (e.g., onset of bobbing followed by pendular oscillations [6]), the fact that the downward movements are faster in both disorders gives rise to a question—are these two distinct eye movement disorders, or do they represent variants of the same underlying disorder? In our records, except for the typical-appearing ocular bobbing cycle that occurs at the 11th second, the other large-amplitude downward excursions (at the 19th, 22nd, and 26th seconds) have features that are unusual for bobbing in that they lack a following tonic interphase interval

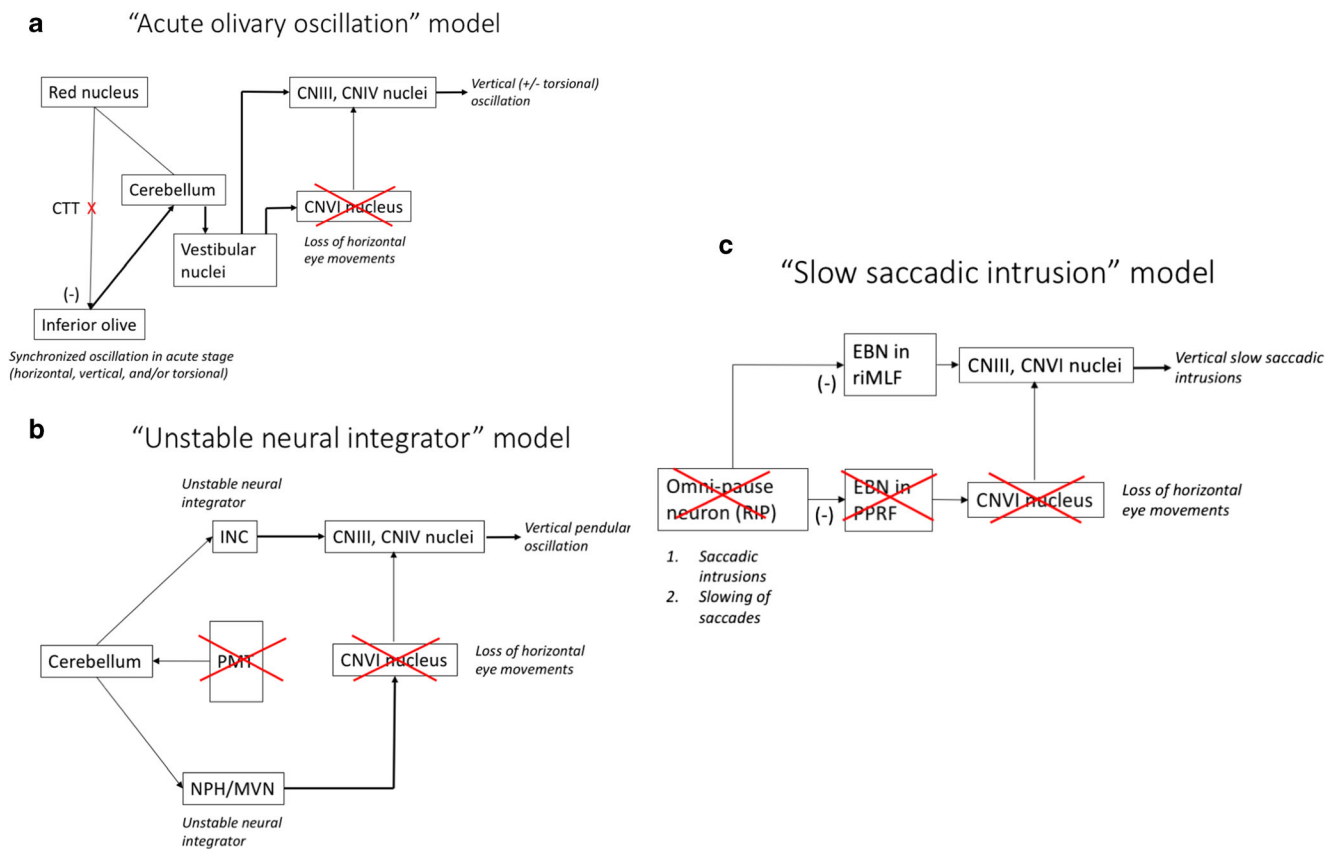


Fig. 6 Three hypothesized models for vertical pendular oscillations and ocular bobbing. **a** “Acute olivary oscillation” model for vertical pendular oscillation (and possibly for ocular bobbing): Disruption of CTT synchronizes the neuronal oscillations in inferior olive and results in pendular ocular oscillations. **b** “Unstable neural integrator” model for vertical pendular oscillation: Damage to pontine cell groups of the PMT causes an unstable neural integrator, which controls horizontal (i.e., NPH and MVN) and vertical gaze holding (i.e., INC), and results in pendular oscillations. **c** “Slow saccadic intrusion” for ocular bobbing (and possibly

for vertical pendular oscillation): Damage to pontine omnipause neurons results in slow saccadic intrusions. The oscillations are mainly vertical since the PPRF and/or abducens nucleus—which control horizontal eye movements—are involved so that horizontal gaze palsies are common. CTT, central tegmental tract; PMT, paramedian tract; NPH, nucleus prepositus hypoglossi; MVN, medial vestibular nucleus; INC, interstitial nucleus of Cajal; RIP, nucleus of raphe interpositus; EBN, excitatory burst neuron; riMLF, rostral interstitial nucleus of medial longitudinal fasciculus

of relative stability of fixation, and their frequencies are higher. Instead, the appearance of these “transitional” waveforms lies between typical ocular bobbing and typical acute pendular oscillations and suggests that these two types of abnormal eye movements might have a common underlying physiological mechanism.

The cause of ocular bobbing remains elusive. One hypothesis is that because most patients with ocular bobbing have a horizontal gaze paralysis due to pontine stroke, ocular bobbing could represent the release of vertical eye movements [1]. It is thought that the pontine omnipause neurons (located in nucleus raphe interpositus in paramedian pontine reticular formation [PPRF]) help to both start and stop horizontal and vertical saccades. Pontine lesions involving the omnipause neurons have been reported to cause fixation instability in the form of macrosaccadic oscillations—large amplitude, back and forth, saccades with an intersaccadic interval of several hundred milliseconds [24]. Slow saccades are also reported after lesions of omnipause neurons [25, 26], even though

there are still superimposed, small amplitude, high-frequency oscillations. In our case, the ocular bobbing at the 11th second (Fig. 4) contains a tonic phase interval of nearly complete steady fixation lasting 0.5 s, although this is a little longer than the usual intersaccadic intervals of macrosaccadic oscillations [20]. Even so, it still seems possible that the ocular bobbing might reflect abnormal saccade behavior resulting from damage to omnipause neurons. These saccadic oscillations could be pure vertical because horizontal oscillations would be blocked by simultaneous involvement of the PPRF (excitatory burst neurons) and/or abducens nuclei due to pontine hemorrhage.

With normal saccades, the peak velocity increases with the amplitude. In the acute vertical pendular oscillations recorded in our patient, this correlation also held true. This raises the possibility that acute pendular oscillations and ocular bobbing could simply represent a series of vertical saccades, albeit slower than normal owing to omnipause involvement. However, given the sequential

appearance of ocular bobbing, acute pendular oscillations, and finally OPT in some patients, another idea is that bobbing could be an immediate manifestation of bilateral CTT damage [6]. In this hypothesis, ocular bobbing, acute pendular oscillations, and OPT could be manifestations of CTT interruption at different stages, which could relate to how and when the cerebellum modulates the inferior olivary outputs [7]. While synchronous discharge of the inferior olive is periodic and jerky [27], cerebellar plasticity may be responsible for aperiodic and smooth discharges [7]. This mechanism may partially explain the transition from ocular bobbing (intermittent and jerky) to pendular oscillation (continuous but less symmetric), and finally to OPT (continuous and symmetric).

While we recognize these ideas are speculative, they still provide a roadmap for further study of the dramatic eye movement disorders shown by these patients in the acute phases after brainstem stroke. Sequential quantitative studies of the evolution of the “bobbing” and the “pendular” components of the oscillations in the first days and weeks after the initial ictus might provide the necessary information to decipher the complex patterns of changes in ocular motor control associated with acute lesions of the pons.

Figure 6 a–c summarize the three hypothesized models and possible anatomical correlates for the mechanisms of vertical pendular oscillation and/or ocular bobbing after pontine hemorrhage—“acute olivary oscillation,” “unstable neural integrator,” and “slow saccadic intrusion”. In the future, longitudinal VOG recordings of the dynamic change of ocular bobbing and vertical pendular oscillations are required to further unravel the mystery of the complexities of ocular bobbing, pendular oscillations, and OPT.

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Data Availability Statement All available data can be obtained by contacting the corresponding author.

Author Contributions Tzu-Pu Chang collected the data, interpreted the data, drafted the manuscript and revised the manuscript.

Daniel R. Gold interpreted the data, drafted the manuscript and revised the manuscript.

Jorge Otero-Millan analyzed and interpreted the data.

Bor-Ren Huang collected the data.

David S. Zee interpreted the data, drafted the manuscript and revised the manuscript.

Compliance with Ethical Standards

The study was performed in accordance with the ethical standards laid down in the Declaration of Helsinki (1964).

Informed Consent The informed and written consent for publication of the data and videos was obtained by the patient’s family.

Conflict of Interests The authors declare that they have no conflict of interest.

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